Predicting Adverse Outcomes and 30-Day Readmission in Emergency Departments Using Machine Learning

Sultan Sarıbaş

Department of Artificial Engineering
Gebze Technical University
Gebze, Kocaeli
s.saribas2026@gtu.edu.tr

***Abstract*— Emergency departments require rapid and accurate clinical decision support to identify high-risk patients early and improve resource allocation. This project proposes a machine learning framework for predicting two clinically important outcomes from emergency department records in Türkiye: adverse outcomes and 30-day readmission. The study will use the public Kaggle dataset from the "AI in Emergency Medicine - Turkiye" competition, which contains one million emergency department encounters collected between January 2019 and December 2024 and represented by 135 features [1]. The project will compare several supervised learning algorithms, including logistic regression, random forest, and gradient boosting-based methods, and evaluate them using classification metrics such as ROC-AUC, F1-score, precision, and recall. The expected outcome is a comparative analysis of predictive models and an interpretable machine learning pipeline for emergency medicine risk prediction.**

***Keywords*—Emergency Medicine, Adverse Outcome Prediction, Clinical Risk Prediction, Readmission Prediction, Clinical Decision Support Systems**

I. Introduction

Emergency medicine is one of the most time-critical domains in healthcare, where early risk estimation can directly affect patient outcomes and hospital efficiency. Artificial intelligence has recently gained increasing attention in emergency care for triage, diagnosis, and outcome prediction tasks. Recent reviews report that AI applications in emergency medicine show strong potential, especially in structured data mining, risk prediction, and clinical decision support, while also highlighting the need for larger-scale and better-generalized studies [2].

This project focuses on predicting adverse outcomes and 30-day readmission for emergency department encounters. These are important because both deterioration risk and short-term revisit risk can help clinicians prioritize patients, optimize interventions, and improve patient management. The selected dataset is particularly suitable for this task because it is large-scale, multimodal, and directly designed around emergency medicine outcome prediction [1].

The main objective of this project is to build and compare machine learning models that can estimate these risks from emergency department encounter data. The project also aims to identify which methods are most suitable for this type of structured clinical prediction problem.

II. Literature Review

An easy way to comply with the conference paper formatting requirements is to use this document as a template and simply type your text into it.

Artificial intelligence research in emergency medicine has expanded rapidly, with applications spanning image processing, signal processing, text mining, and structured tabular prediction. A 2025 scoping review found that structured-data ML models are widely used in emergency medicine for prediction tasks such as triage support, risk analysis, and outcome prediction, but emphasized that data quality, preprocessing, external validation, and interpretability remain major challenges [2].

Several prior studies have shown that machine learning can support emergency department decision-making. Feretzakis et al. investigated hospital admission prediction in emergency departments using common clinical biomarkers and demographic information, showing that machine learning models can produce useful predictive performance for real-world ED settings [3]. In another recent benchmark, MDS-ED demonstrated that multimodal emergency care data can be effectively used to predict deterioration-related outcomes, with strong AUROC performance on several critical targets such as ICU admission and mortality [4].

These studies suggest that clinical outcome prediction in emergency medicine is a promising ML application area. However, many studies are limited by dataset scale, modality scope, or deployment realism. This project contributes by applying and comparing practical machine learning models on a large Türkiye-focused emergency department dataset specifically designed for adverse outcome and readmission prediction [1].

III. METHODS

A. Problem Definition

This project addresses a supervised learning problem in emergency medicine. The goal is to predict whether a patient encounter will result in “an adverse outcome”, and/or “a 30-day emergency department readmission”.

Both tasks will be formulated as binary classification problems. Since the dataset is derived from emergency department encounters and the competition framing explicitly targets adverse outcomes and 30-day readmission, the project will center on risk prediction from structured clinical features.

B. Data

The project will use the AI in Emergency Medicine - Türkiye Kaggle competition dataset. According to the competition description, the dataset contains 1,000,000 emergency department encounters, recorded from January 2019 to December 2024, with 135 features describing each encounter [1].

The dataset will be obtained from Kaggle and examined for:

- missing values,
- feature types,
- class imbalance,
- potential outliers,
- data leakage risks,
- train-validation-test splitting strategy.

Preprocessing will include cleaning missing values, encoding categorical variables, scaling numerical variables when needed, and performing feature engineering if justified by exploratory analysis.

C. Methodology

The project will be executed in the following stages:

1) *Exploratory Data Analysis (EDA):* The dataset structure, feature distributions, missingness, and target imbalance will be analyzed.

2) Pre-processing: Categorical encoding, numerical scaling, missing value handling, and possible feature selection will be applied.

3) Baseline Modelling: A simple interpretable baseline such as logistic regression will be implemented first.

4) Advanced Modelling: More powerful tabular machine learning models such as random forest, XGBoost, and LightGBM will be trained and compared.

5) Model Tuning: Hyperparameter optimization will be conducted for the best-performing models.

6) Interpretation and Analysis: Feature importance analysis and error analysis will be performed to better understand model decisions.

This workflow is consistent with common structured-data clinical prediction pipelines and is appropriate for large-scale emergency medicine classification tasks supported by prior literature [2], [3], [4].

D. Techniques

The specific techniques planned for this project are:

- Logistic Regression as a baseline model for interpretability
- Random Forest for nonlinear decision boundaries and robustness
- XGBoost / LightGBM for high-performance gradient boosting on tabular data
- Class imbalance handling through class weighting or resampling methods
- Feature importance analysis using model-based importance scores or SHAP-style interpretation
- Cross-validation for reliable model comparison

These methods are chosen because the dataset is large-scale, tabular, and clinically structured, making tree-based ensemble methods especially suitable. Recent emergency medicine literature also indicates that ML-based structured data models are widely used for outcome prediction tasks [2], [3].

E. Evaluation

Model performance will be evaluated using classification metrics appropriate for potentially imbalanced medical datasets which are ROC-AUC, F1-score, Precision, Recall, Accuracy and, optionally PR-AUC, if class imbalance is strong.

Among these metrics, recall and F1-score will be particularly important because missing high-risk patients may have serious clinical consequences. In addition, model interpretability and robustness will be considered as secondary evaluation criteria, since emergency medicine applications require clinical trust and practical usability. This concern is also emphasized in recent reviews of AI in emergency medicine [2].

F. Expected Deliverables

By the end of the quarter, the project is expected to produce:

- A complete IEEE-format final report
- Cleaned and documented source code
- EDA results and pre-processing documentation
- A comparative evaluation of multiple ML models
- Visualizations of performance metrics and important features

- A short presentation or demo summarizing findings
- Links and Bookmarks

IV. Conclusions

This project is expected to provide a comparative machine learning study for emergency medicine risk prediction using a large-scale Türkiye-based dataset. Its main contribution will be demonstrating how structured clinical data can be used to predict clinically meaningful outcomes such as deterioration risk and readmission. The project may also provide insight into which tabular ML models offer the best balance between predictive performance and interpretability in emergency department settings.

References

[1] "AI in Emergency Medicine - Turkiye," *@kaggle,* 2026. https://www.kaggle.com/competitions/ai-emergency-medicine-turkiye/overview.

[2] G. B. Berikol, A. Kanbakan, B. Ilhan, and F. Doğanay, "Mapping artificial intelligence models in emergency medicine: A scoping review on artificial intelligence performance in emergency care and education," *Turkish Journal of Emergency Medicine*, vol. 25, no. 2, pp. 67–91, Apr. 2025, doi: https://doi.org/10.4103/tjem.tjem_45_25.

[3] G. Feretzakis *et al.*, "Using Machine Learning Techniques to Predict Hospital Admission at the Emergency Department" *arXiv.org*, 2021. https://arxiv.org/abs/2106.12921.

[4] J. M. L. Alcaraz, H. Bouma, and N. Strodthoff, "Enhancing clinical decision support with physiological waveforms — A multimodal benchmark in emergency care," *Computers in Biology and Medicine*, vol. 192, p. 110196, Jun. 2025, doi: https://doi.org/10.1016/j.compbiomed.2025.110196.